# Linear Regression, Regularization

Dr. Ahmed Aljarah

# Correction

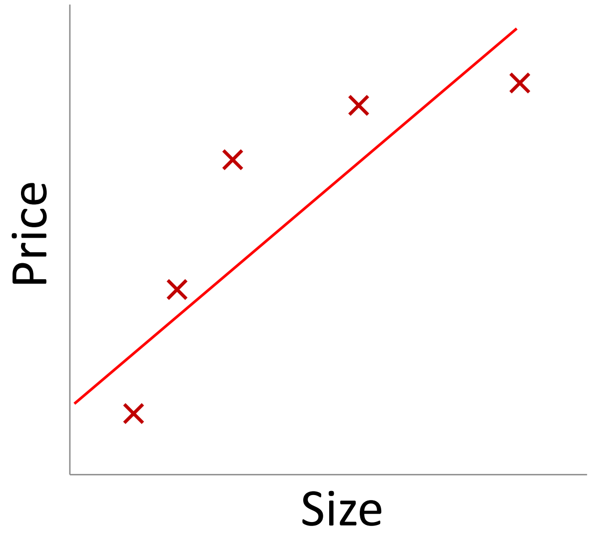
Poor performance occurs for two reasons:

1. Underfitting:
   - Correction: Increase the number of features.

2. Overfitting:
   - Correction 1: Reduce the number of features.

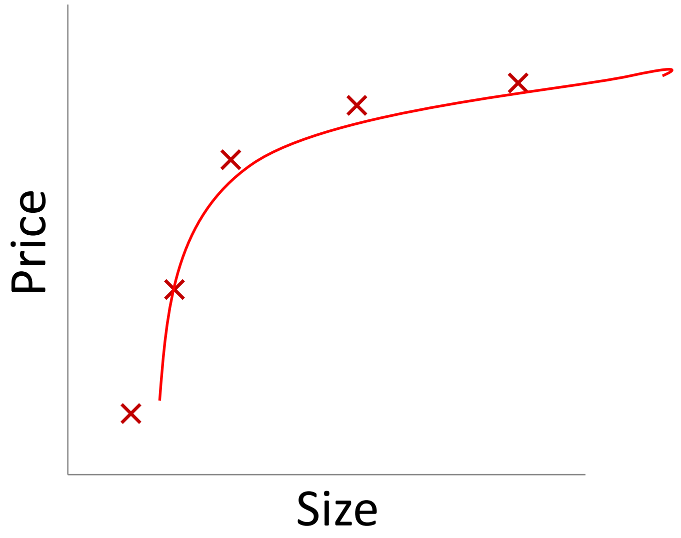
   - Correction 2: Keep all features but reduce the magnitude/value of the parameters $\theta_j$ (regularization).

# Example:



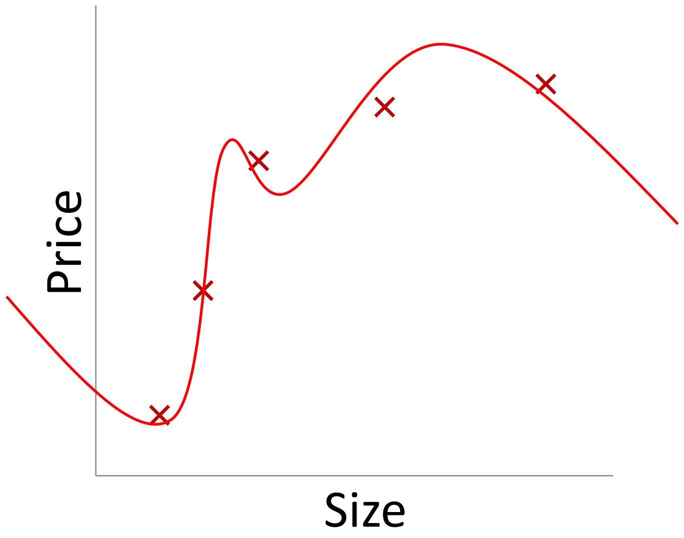

$$\theta_0 + \theta_1 x$$

Underfit




$$\theta_0 + \theta_1 x + \theta_2 x^2$$

Balanced fit




$$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$$

Overfit

# Regularization

- Regularization is a technique used to reduce errors by fitting the function appropriately on a given training set and avoiding overfitting.

- Overfitting occurs when a regression model fits the training data too well and performs poorly on new data.

- To reduce overfitting, we use regularization, which adds a penalty to large model coefficients.

- Three common regularization methods are:
  1. Lasso Regression (L1)
  2. Ridge Regression (L2)
  3. Elastic net

# Lasso regression (L1 regularization)

- Lasso regression adds a penalty based on the absolute values of the coefficients.
- If some coefficients become too large, the model adds a penalty, forcing them to become smaller.

Cost Function: $J(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y_{i\_pred} - y_i)^2 + \lambda\sum_{j=1}^{p} |\theta_j|$

Regulariztion parameter

- Parts of the equation:

  First term of the formula: prediction error.

  - n is the number of data points

  Second term of the formula: penalty based on absolute values of coefficients.

- The goal is to minimize the cost.

# Gradient Descent Optimization for L1

$$\theta_j = \theta_j - \alpha \cdot \frac{\partial J}{\partial \theta_j}$$

$$\frac{\partial j}{\partial \theta_j} = \frac{2}{n}\sum(y_{pred} - y)x_j + \lambda \cdot \text{sign}(\theta_j)$$

$\text{sign}(\theta_j) = 1$ or $0$ or $-1$

- $+1$ if $\theta_j > 0$
- $-1$ if $\theta_j < 0$
- $0$ if $\theta_j = 0$

Final update

$$\theta_j = \theta_j - \alpha \cdot \left[\frac{2}{n}\sum(y_{pred} - y)x_j + \lambda \cdot \text{sign}(\theta_j)\right]$$

Imagine:

- Cost function = height of a hill
- Gradient = slope of the hill
- Update rule = how you walk downhill

An important feature of Lasso is that it can reduce some coefficients to exactly zero, which means it can automatically select important features.

# Example

Suppose:

- $\theta = 3$
- $\lambda = 1$
- $\alpha = 0.1$
- Error part = 4

- Step 1: sign(3) =1
- Step 2: $\frac{\partial j}{\partial \theta_j} = \frac{2}{n}\sum(y_{pred} - y)x_j + \lambda \cdot \mathrm{sign}(\theta_j)$ -> 4+1 = 5
- Step 3: $\theta_j = \theta_j - \alpha \, . \, \frac{\partial J}{\partial \theta_j}$ ->$\theta$=3−0.1(5)=2.5 -> $\theta$ decreased

# Ridge regression (L2 regularization)

- Ridge regression adds a penalty based on the squared values of the coefficients.
- It prevents coefficients from becoming too large but does not force them to zero.

Cost Function: $J(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y_{i-predict} - y_i)^2 + \lambda\sum_{j=1}^{p}\theta_j^2$

- Parts of the equation:
  - First term: prediction error
  - Second term: penalty using squared coefficients

- Ridge regression works well when predictor variables are highly correlated.

# Gradient Descent Optimization for L2

$$\theta_j = \theta_j - \alpha \cdot \frac{\partial J}{\partial \theta_j}$$

$$\frac{\partial j}{\partial \theta_j} = \frac{2}{n} \sum (y_{pred} - y) x_j + 2\lambda\theta_j$$

Final update

$$\theta_j = \theta_j - \alpha \cdot [\frac{2}{n} \sum (y_{pred} - y) x_j + 2\lambda\theta_j]$$

Ridge Regularization:

- Reduces coefficient size, prevents instability
- Prevents extreme weights, avoids overfitting
- Simplifies the model, improves generalization

# Elastic net regression

- Elastic Net combines Lasso and Ridge penalties.
- It uses both L1 and L2 regularization at the same time.

Cost Function: $J(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y_{i-predict} - y_i)^2 + \lambda\sum_{j=1}^{p} |\theta_j| + \frac{1}{2}(1-\alpha)\lambda\sum_{j=1}^{p}\theta_j^2$

Parameters

- λ (lambda): controls how strong the penalty is
- α (alpha): controls the mix between Lasso and Ridge

If:

α = 1 → Lasso
α = 0 → Ridge
0 < α < 1 → mixture of both(Elastic net)

# Gradient Descent Optimization for L1+L2

$\theta_j = \theta_j - \eta . \frac{\partial J}{\partial \theta_j}$

$\frac{\partial j}{\partial \theta_j} = \frac{2}{n} \sum (y_{pred} - y) x_j + \lambda.\text{sign}(\theta_j) + (1- \alpha)\lambda\theta_j$

Final update

$\theta_j = \theta_j - \eta \ . \ [\frac{2}{n} \sum (y_{pred} - y) x_j + \lambda.\text{sign}(\theta_j) + (1- \alpha)\lambda\theta_j]$

# Comparison between regularization techniques

| Technique | Penalty | Key Feature |
| --- | --- | --- |
| **Lasso** | Absolute values (L1) | Can remove variables |
| **Ridge** | Squared values (L2) | Handles correlated variables |
| **Elastic Net** | L1 + L2 | Combines both advantages |

- Lasso: selects important features
- Ridge: stabilizes coefficients
- Elastic Net: combines both methods

# Example 2:

## Implement a linear regression model with L1 and L2 regularization using python

Melbourne

| Suburb | Address | Rooms | Type | Price | Method | SellerG | Date | Distance | Postcode | Bedroom2 | Bathroom | Car | Landsize | BuildingArea | YearBuilt | CouncilArea | Lattitude | Longtitude | Regionname | Propertycount |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Abbotsford | 68 Studley St | 2 | h | | SS | Jellis | 3/09/2016 | 2.5 | 3067 | 2 | 1 | 1 | 126 | | | Yarra City Council | -37.8014 | 144.9958 | Northern Metropolitan | 4019 |
| Abbotsford | 85 Turner St | 2 | h | 1480000 | S | Biggin | 3/12/2016 | 2.5 | 3067 | 2 | 1 | 1 | 202 | | | Yarra City Council | -37.7996 | 144.9984 | Northern Metropolitan | 4019 |
| Abbotsford | 25 Bloomburg St | 2 | h | 1035000 | S | Biggin | 4/02/2016 | 2.5 | 3067 | 2 | 1 | 0 | 156 | 79 | 1900 | Yarra City Council | -37.8079 | 144.9934 | Northern Metropolitan | 4019 |
| Abbotsford | 18/659 Victoria St | 3 | u | | VB | Rounds | 4/02/2016 | 2.5 | 3067 | 3 | 2 | 1 | 0 | | | Yarra City Council | -37.8114 | 145.0116 | Northern Metropolitan | 4019 |
| Abbotsford | 5 Charles St | 3 | h | 1465000 | SP | Biggin | 4/03/2017 | 2.5 | 3067 | 3 | 2 | 0 | 134 | 150 | 1900 | Yarra City Council | -37.8093 | 144.9944 | Northern Metropolitan | 4019 |
| Abbotsford | 40 Federation La | 3 | h | 850000 | PI | Biggin | 4/03/2017 | 2.5 | 3067 | 3 | 2 | 1 | 94 | | | Yarra City Council | -37.7969 | 144.9969 | Northern Metropolitan | 4019 |
| Abbotsford | 55a Park St | 4 | h | 1600000 | VB | Nelson | 4/06/2016 | 2.5 | 3067 | 3 | 1 | 2 | 120 | 142 | 2014 | Yarra City Council | -37.8072 | 144.9941 | Northern Metropolitan | 4019 |
| Abbotsford | 16 Maugie St | 4 | h | | SN | Nelson | 6/08/2016 | 2.5 | 3067 | 3 | 2 | 2 | 400 | 220 | 2006 | Yarra City Council | -37.7965 | 144.9965 | Northern Metropolitan | 4019 |
| Abbotsford | 53 Turner St | 2 | h | | S | Biggin | 6/08/2016 | 2.5 | 3067 | 4 | 1 | 2 | 201 | | 1900 | Yarra City Council | -37.7995 | 144.9974 | Northern Metropolitan | 4019 |
| Abbotsford | 99 Turner St | 2 | h | | S | Collins | 6/08/2016 | 2.5 | 3067 | 3 | 2 | 1 | 202 | | 1900 | Yarra City Council | -37.7996 | 144.9989 | Northern Metropolitan | 4019 |
| Abbotsford | 129 Charles St | 2 | h | 941000 | S | Jellis | 7/05/2016 | 2.5 | 3067 | 2 | 1 | 0 | 181 | | | Yarra City Council | -37.8041 | 144.9953 | Northern Metropolitan | 4019 |
| Abbotsford | 124 Yarra St | 3 | h | 1876000 | S | Nelson | 7/05/2016 | 2.5 | 3067 | 4 | 2 | 0 | 245 | 210 | 1910 | Yarra City Council | -37.8024 | 144.9993 | Northern Metropolitan | 4019 |
| Abbotsford | 121/56 Nicholson St | 2 | u | | PI | Biggin | 7/11/2016 | 2.5 | 3067 | 2 | 2 | 1 | 4292 | 82 | 2009 | Yarra City Council | -37.8078 | 144.9965 | Northern Metropolitan | 4019 |
| Abbotsford | 17 Raphael St | 4 | h | | W | Biggin | 7/11/2016 | 2.5 | 3067 | 6 | 2 | 0 | 230 | 147 | 1860 | Yarra City Council | -37.8066 | 144.9936 | Northern Metropolitan | 4019 |
| Abbotsford | 98 Charles St | 2 | h | 1636000 | S | Nelson | 8/10/2016 | 2.5 | 3067 | 2 | 1 | 2 | 256 | 107 | 1890 | Yarra City Council | -37.806 | 144.9954 | Northern Metropolitan | 4019 |
| Abbotsford | 217 Langridge St | 3 | h | 1000000 | S | Jellis | 8/10/2016 | 2.5 | 3067 | | | | | | | Yarra City Council | | | Northern Metropolitan | 4019 |
| Abbotsford | 18a Mollison St | 2 | t | 745000 | S | Jellis | 8/10/2016 | 2.5 | 3067 | | | | | | | Yarra City Council | | | Northern Metropolitan | 4019 |
| Abbotsford | 6/241 Nicholson St | 1 | u | 300000 | S | Biggin | 8/10/2016 | 2.5 | 3067 | 1 | 1 | 1 | 0 | | | Yarra City Council | -37.8008 | 144.9973 | Northern Metropolitan | 4019 |
| Abbotsford | 10 Valiant St | 2 | h | 1097000 | S | Biggin | 8/10/2016 | 2.5 | 3067 | 3 | 1 | 2 | 220 | 75 | 1900 | Yarra City Council | -37.801 | 144.9989 | Northern Metropolitan | 4019 |
| Abbotsford | 403/609 Victoria St | 2 | u | 542000 | S | Dingle | 8/10/2016 | 2.5 | 3067 | | | | | | | Yarra City Council | | | Northern Metropolitan | 4019 |
| Abbotsford | 2 Rich St | 2 | h | | SP | Biggin | 10/12/2016 | 2.5 | 3067 | 2 | 1 | 1 | 176 | 80 | 1925 | Yarra City Council | -37.7998 | 144.9972 | Northern Metropolitan | 4019 |
| Abbotsford | 25/84 Trenerry Cr | 2 | u | 760000 | SP | Biggin | 10/12/2016 | 2.5 | 3067 | | | | | | | Yarra City Council | | | Northern Metropolitan | 4019 |
| Abbotsford | 106/119 Turner St | 1 | u | 481000 | SP | Purplebricks | 10/12/2016 | 2.5 | 3067 | | | | | | | Yarra City Council | | | Northern Metropolitan | 4019 |
| Abbotsford | 411/8 Grosvenor St | 2 | u | 700000 | VB | Jellis | 12/11/2016 | 2.5 | 3067 | 2 | 2 | 1 | 0 | | | Yarra City Council | -37.811 | 145.0067 | Northern Metropolitan | 4019 |
| Abbotsford | 40 Nicholson St | 3 | h | 1350000 | VB | Nelson | 12/11/2016 | 2.5 | 3067 | 3 | 2 | 2 | 214 | 190 | 2005 | Yarra City Council | -37.8085 | 144.9964 | Northern Metropolitan | 4019 |
| Abbotsford | 123/56 Nicholson St | 2 | u | 750000 | S | Biggin | 12/11/2016 | 2.5 | 3067 | 2 | 2 | 1 | 0 | 94 | 2009 | Yarra City Council | -37.8078 | 144.9965 | Northern Metropolitan | 4019 |

```python
# Importing the libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
# Connect colab with dirve
from google.colab import drive
drive.mount('/content/drive')
```

```python
#Importing the dataset
df = pd.read_csv("/content/drive/ MyDrive/AI course/ Melbourne.csv")
pd.set_option("display.max_columns",None)
df.head()
df.shape
df.info()
df.size
df.nunique()
```

```
# filtering columns
column_to_use=['Suburb', 'Rooms', 'Type', 'Method', 'SellerG', 'Regionname', 'Propertycount',
'Distance', 'CouncilArea', 'Bedroom2', 'Bathroom', 'Car', 'Landsize', 'BuildingArea','Price']
df = df[column_to_use]
df.head()
df.shape
```

```
# filtering columns
df.isna().sum()
```

```
# Some feature's missing values can be treated as zero
# like 0 for Propertycount, Bedroom2 will refer to other class of NA values
# like 0 for Car feature will mean that there's no car parking feature with house
cols_to_fill_zero = ['Propertycount', 'Distance', 'Bedroom2', 'Bathroom', 'Car']
df[cols_to_fill_zero]=df[cols_to_fill_zero].fillna(0)
df.isna().sum()
```

```
# other continuous features can be imputed with mean for faster results since our focus is on
#Reducing overfitting using Lasso and Ridge Regression
df["Landsize"] = df["Landsize"].fillna(df["Landsize"].mean())
df["BuildingArea"] = df["BuildingArea"].fillna(df["BuildingArea"].mean())
df.isna().sum()
```

```
df.dropna(inplace=True)
df.isna().sum()
df.head()
df.shape
```

```python
# transform text columns to numerical
df=pd.get_dummies(df,drop_first=True)
df.shape
```

```python
#Extracting the Independent and Dependent variables
x = df.drop("Price",axis=1)
y = df["Price"]
```

```python
# Splitting the dataset into the Training set and Test set
from sklearn.model_selection import train_test_split
x_trian, x_test, y_trian,y_test = train_test_split(x,y,test_size=0.3,random_state=2)
```

```python
# Create the Linear Regrission model
from sklearn.linear_model import LinearRegression
lr = LinearRegression()
lr.fit(x_trian,y_trian)
```

```python
# Calculating the R squared value
y_predict = lr.predict(x_test)
print(y_predict)
from sklearn.metrics import r2_score
r2_score(y_test,y_predict)
```

```python
# Calculating the Coefficients and intercept
print(lr.coef_)
print(lr.intercept_)
```

```python
# create L1 (lasso)
from sklearn.linear_model import Lasso
las = Lasso(alpha=50,max_iter=100, tol=0.1 )
las.fit(x_trian,y_trian)
```

```python
# create L2 (Ride)
from sklearn.linear_model import Ridge
r = Ridge(alpha=50,max_iter = 100, tol=0.1)
r.fit(x_trian, y_trian)
```

```python
# create L2 +L2 (Elastic net)
from sklearn.linear_model import ElasticNet
elastic_net_model = ElasticNet(alpha=50, l1_ratio=0.5, random_state=0)
elastic_net_model.fit(x_trian, y_trian)
```